

MODULE 2

Embedded Hardware: RAK4631 WisBlock

TinyML — Condition Monitoring on Microcontrollers

Day 1 · Module 2 of 9

Module 2 = the embedded half of TinyML

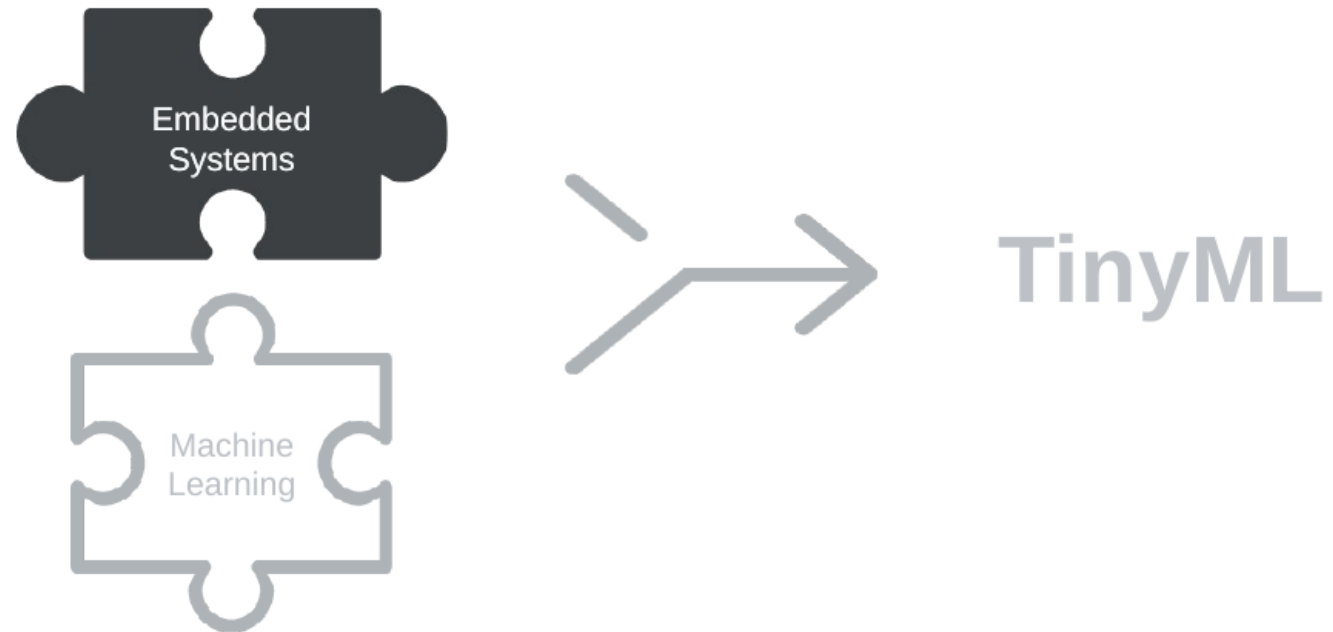


Figure: TinyML on edX (HarvardX), V.J. Reddi et al. — used with permission

The brain: nRF52840 (on the RAK4631 core)

Spec	Value	TinyML relevance
CPU	Cortex-M4F @ 64 MHz	hardware float — features & NN math
RAM	256 KB	model + buffers + stack live here
Flash	1 MB	code + model weights
PDM peripheral	✓	digital microphones, zero CPU cost
EasyDMA		peripherals move data while CPU sleeps
Radio	BLE 5 (+ SX1262 LoRa on module)	transmit <i>results</i> , not raw data

RAM vs flash — the two budgets

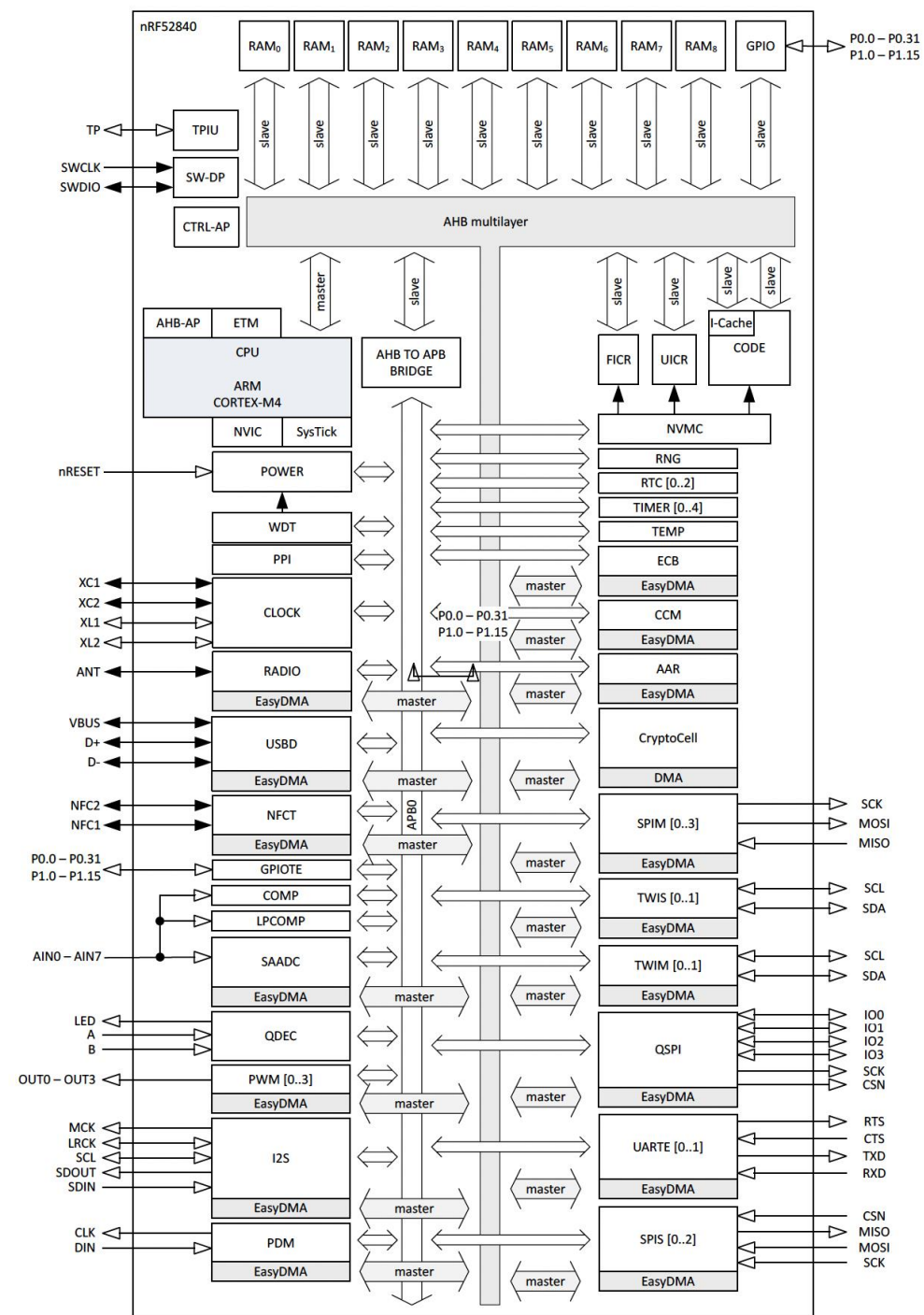
- **Flash (1 MB):** your code + the BSP + **model weights**. Written at upload.
- **RAM (256 KB):** stack, heap, sensor buffers, **model activations**.

Rules of thumb:

- 2 s of 16 kHz 16-bit audio = **64 KB RAM** — a quarter of everything
- 10k-param float32 model = 40 KB flash + activations in RAM
- Peak RAM usage happens *inside* inference — not visible in the map file

nRF52840

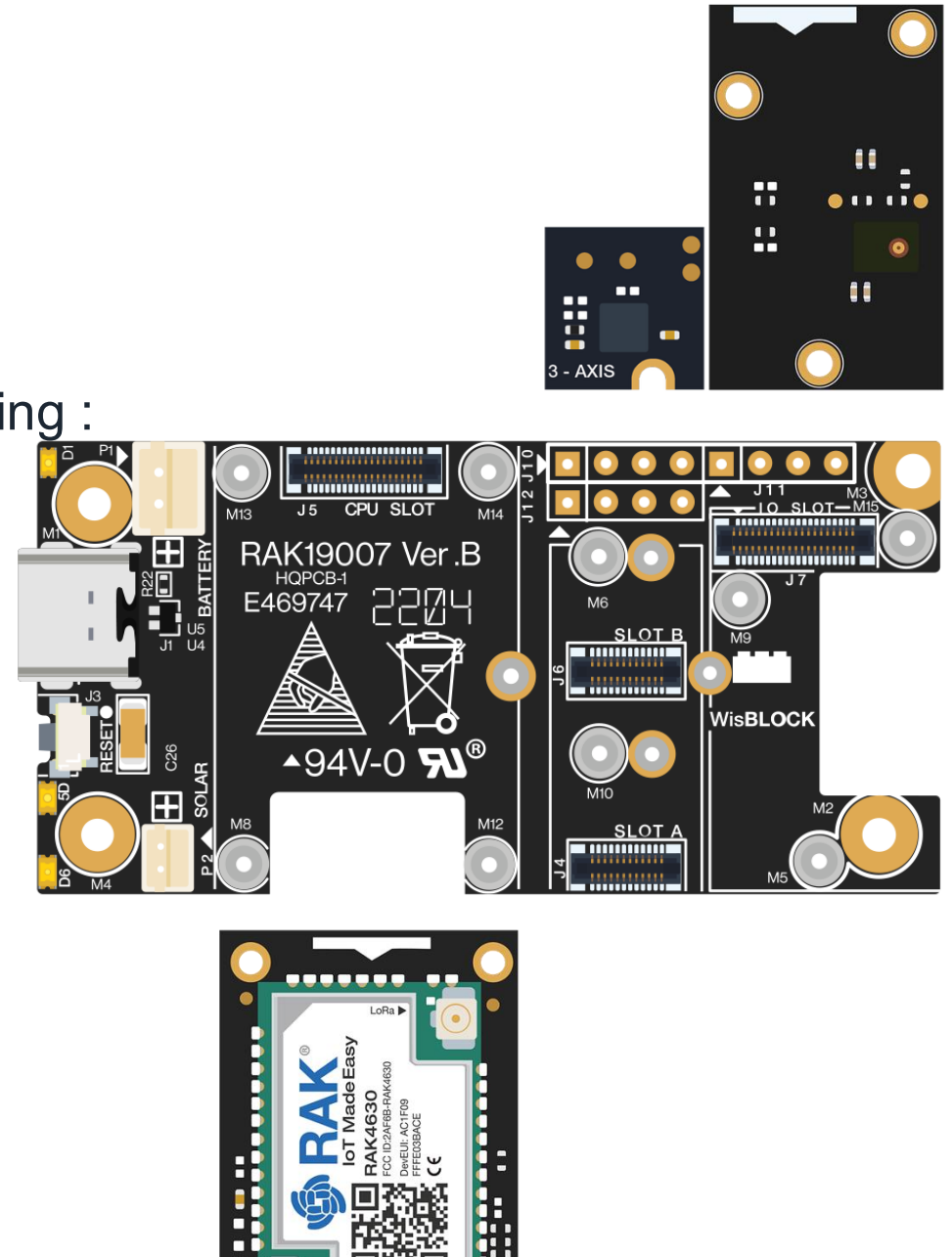
Popular Cortex M4F
Builtin BLE / 2.4GHZ radio
Plenty of peripherals
DMA + Multilayer AHB



WisBlock: embedded Lego

A modular system — no soldering, no breadboard, no wiring :

- **Core** module — CPU + radio (our RAK4631)
- **Base** board — power, USB, slots (our RAK19007)
- **Sensor** modules — small, I²C/SPI, slots A–D
- **IO** modules — larger, more pins (our microphone)



Your kit

Part	What	Where
RAK4631	nRF52840 + SX1262 core	core slot
RAK19007	base board, USB-C	—
RAK1904	LIS3DH 3-axis accelerometer	sensor slot A
RAK18000	2× MP34DT06J PDM microphones	IO slot
120 mm PC fan	your "pump"	-

Assembly rules

1. **USB unplugged** while adding/removing modules
2. Align the connector **and** the mounting-screw hole — the module only fits one way
3. **Never force it.** If it needs force, it's the wrong slot or wrong orientation

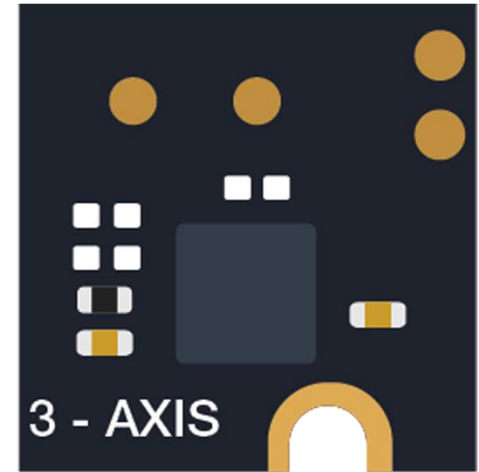
Then: screw the modules down. Vibration sensing on a machine that vibrates — loose modules make their own dataset.

RAK1904 — the accelerometer

- ST **LIS3DH**, 3-axis MEMS accelerometer
- I²C address **0x18**, interrupt line on `WB_IO5`
- Range: $\pm 2/4/8/16$ g — we use **± 4 g**
- Data rate: 1 Hz – 5.3 kHz — we use **50–100 Hz**
- Driver: **Adafruit LIS3DH** Arduino library (plus BusIO + Unified Sensor)

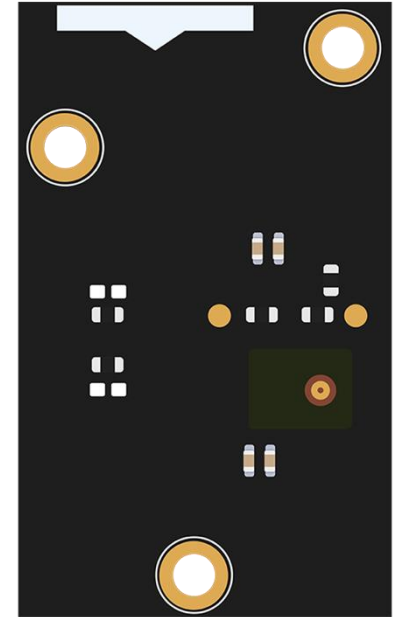


Gotcha: sensor slots are power-gated — drive `WB_IO2` **HIGH** first!



RAK18000 — the microphone

- 2× ST **MP34DT06J** PDM MEMS microphones (stereo — we use mono)
- Sits in the **IO slot**
- **PDM DATA** = `WB_IO3` , **PDM CLK** = `WB_IO4`
- 62.6 dB SNR, AOP 122.5 dBSPL — plenty for machinery

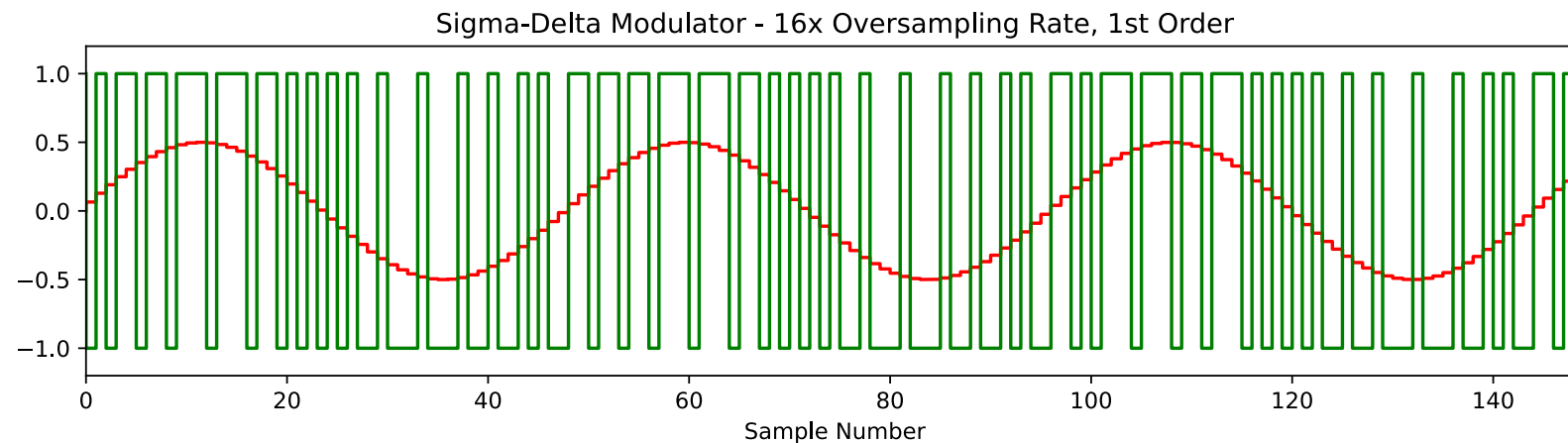


PDM — how a digital microphone talks

- Mic outputs a **1-bit stream** at MHz rate: density of 1s $\propto$ sound pressure
- No analog signal path, no ADC, cheap and noise-immune
- The nRF52840 **PDM peripheral** filters + decimates in hardware:

1-bit @ 1.28 MHz — $\div 80$ (hardware) —► 16-bit PCM @ 16 kHz

- CPU cost: **zero**. Samples appear in RAM via EasyDMA.



Toolchain: VS Code + PlatformIO

- **PlatformIO** — package manager + build system for ~1500 boards
- Reproducible: `platformio.ini` pins platform, board, libraries
- One quirk: RAK4631 isn't in the official registry → **RAK patch** (you did this in the setup guide)

```
[env:wiscore_rak4631]
platform = nordicnrf52
board    = wiscore_rak4631
framework = arduino
monitor_speed = 115200
```

Anatomy of platformio.ini

```
[env:wiscore_rak4631]
platform = nordicnrf52      ; toolchain + chip support
board    = wiscore_rak4631  ; pin map, memory layout (from RAK patch)
framework = arduino         ; Adafruit nRF52 core
monitor_speed = 115200      ; serial monitor baud

lib_deps =                  ; auto-installed, version-locked
  adafruit/Adafruit LIS3DH
  adafruit/Adafruit BusIO
  adafruit/Adafruit Unified Sensor
```

Arduino on nRF52: what the BSP gives you

- `setup()` / `loop()` , `digitalWrite` , `Wire` (I²C), `SPI` , `Serial` (USB CDC)
- WisBlock pin names: `WB_IO1..7` , `LED_GREEN` , `LED_BLUE`
- FreeRTOS underneath (`delay()` actually sleeps — low power for free)
- Bundled **nrfx** drivers for bare-metal peripherals — that's our PDM path

Flashing & the bootloader dance

- Upload = `adafruit-nrfutil` over USB serial (automatic)
- Board stuck / no port? **Double-press RESET** → green LED breathes, `RAK4631` USB drive appears = bootloader mode → upload again
- Serial monitor: **115200 baud**. Note: USB serial re-enumerates after reset — give it a second, reopen the monitor



Lab checkpoint 1 — blink (~15 min)

[exercises/module2-hardware-bringup/blink/](#)

- Build → upload → green LED blinks
- Upload fails? Double-press RESET (bootloader mode) and retry
- **Done when:** you've changed the blink rate and re-flashed — the loop works

Serial: your debugger, plotter, and data pipe

- `Serial.begin(115200)` + `Serial.print` — you know this
- PlatformIO monitor: the terminal icon, or `pio device monitor`
- Budget check: 115200 baud $\approx$ **11.5 KB/s** — remember this number in Module 3
- Tip: `while (!Serial) delay(10);` waits for the monitor (but blocks standalone boots!)

I²C on WisBlock

- `Wire` = the sensor-slot I²C bus
- RAK1904 answers at I²C address **0x18**
- Checklist: module seated? slot A? `WB_IO2` **HIGH?**

```
pinMode(WB_IO2, OUTPUT);  
digitalWrite(WB_IO2, HIGH); // power the sensor slots  
delay(100);                // sensor boot time  
Wire.begin();
```



Lab checkpoint 2 — accelerometer (~20 min)

[exercises/module2-hardware-bringup/accel-read/](#)

- Power off → mount RAK1904 in **sensor slot A** → power on
- Build, upload, open serial monitor: live x/y/z at 115200
- **Done when:** $z \approx 9.8 \text{ m/s}^2$ flat on the desk, and it moves when you do



Lab checkpoint 3 — microphone (~20 min)

[exercises/module2-hardware-bringup/mic-level/](#)

- Power off → mount RAK18000 in the **IO slot** → power on
- Build, upload: live RMS level meter on the serial monitor
- Silence? Flip `NRF_PDM_EDGE_LEFTRISING` ↔ `LEFTFALLING` and re-flash
- **Done when:** the number jumps when you talk at the board

Common bring-up failures

Symptom	Likely cause
wiscore_rak4631: unknown board	RAK patch not applied to this PIO install
Upload timeout / no port	bootloader dance: double-press RESET
Failed to find LIS3DH	WB_IO2 not HIGH · wrong slot · not seated
Mic reads all zeros	DATA/CLK swapped (WB_IO3 / WB_IO4)
Serial monitor silent	wrong baud · monitor opened before re-enumeration

Module 2 wrap-up

- ✓
 - Toolchain proven on every laptop
- ✓
 - nRF52840: 256 KB RAM, FPU, PDM + EasyDMA — and what that means
- ✓
 - Accelerometer streaming live data
- ✓
 - Microphone alive and measuring

Next: stop watching numbers scroll by — start **collecting datasets**. Edge Impulse enters the chat.